PAPER_698: Einstein Ring GAL-CLUS-022058s: Gravitational Lensing and UQFF Deflection

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Abstract

GAL-CLUS-022058s (the 'Molten Ring') is a near-complete Einstein ring formed by a galaxy cluster lens at $z_L \sim 0.5$ bending light from a source galaxy at $z_S \sim 1.2.$ The UQFF framework adds an aether correction to the deflection angle.

1. Parameters

Parameter	Value
M_{lens}	$10^{15} M_\odot$ (cluster)
D_L	500 Mpc
D_S	1.5 Gpc
θ_E	10 arcsec

2. Einstein Radius

$$R_E = D_L \sqrt{\frac{4GM_L D_{LS}}{c^2 D_L D_S}}$$

3. Magnification

$$\mu = \frac{u^2+2}{u\sqrt{u^2+4}}, \quad u = \frac{\theta}{\theta_E}$$

4. UQFF Deflection Angle

$$\hat{\alpha}_{UQFF}(r) = \frac{4GM_L}{c^2 r}\left(1 + \frac{\rho_{SCm}}{\rho_{UA}}\right)$$

The aether correction increases deflection by factor $(1 + 0.1) = 1.1$ at UQFF scale.

5. Convergence

$$\kappa = \frac{\Sigma}{\Sigma_{crit}}, \quad \Sigma_{crit} = \frac{c^2 D_S}{4\pi G D_L D_{LS}}$$

References

- Schwab et al. (2010); Hubble ACS; UQFF Framework, Star-Magic Session 174

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